## Subjective randomness: which of two coin-flip sequences looks more random?

`personal_pattern_detection_gain`

```
for each person i:                                          ← person-level parameters
  θᵢ  ideal switching rate;  βᵢ  sensitivity to it
  γᵢ  weight on H/T imbalance;  δᵢ  on the longest streak
  gᵢ  pattern-detection gain (median 1)                     ← one trait scales every structural cue
  λᵢ  lapse (guessing) rate
for each sequence x of length n:
  sw(x)   = share of adjacent flips that switch
  imb(x)  = |#H − #T| / n
  run(x)  = (longest streak − 1) / (n − 1)
  end(x)  = (final streak − 1) / (n − 1)                    ← recency: a long final streak
  pal(x)  = 1 if x reads the same reversed                  ← designed-looking sequences penalised
  anti(x) = 1 if x equals its reversed complement
  var(x)  = variety of run lengths (0 to 1)                 ← varied runs look random
  lop(x)  = (range of the H−T tally − 1) / (n − 1)          ← the most lopsided stretch
  struct(x) = −κ end − ρ pal − χ anti + ω var − η lop       ← cue weights shared by everyone
  dev = sw(x) − θᵢ
  score(x) = −βᵢ dev² · (φ if dev > 0 else 1)               ← over-alternation weighted by a fitted φ
             − γᵢ imb(x) − δᵢ run(x) + gᵢ struct(x)
P(choose a over b) =
    λᵢ/2 + (1 − λᵢ) σ(score(a) − score(b))                  ← a person guesses at rate λᵢ
```

## RSA reference games: which object does the word refer to?

`boundary_avoidable_ambiguity_listener`

```
prior(o) ∝ exp(w·salience(o))                               ← unique, distinctive, many-featured, coloured, typical
L0(o | u) ∝ lex(u, o)                                       ← literal listener
S1(u | o) ∝ lex(u, o) · exp(α [L0(o|u)
        − w_int · interior(u, o)                            ← interior: neither the simplest nor the richest match
        − c_av · avoidable(u, o)]                           ← avoidable: u is ambiguous though o had a unique word
        − w_conf · confusion(u, o))                         ← confusion: u also fits objects that resemble o
L1(o | u) ∝ prior(o) · S1(u | o)
S2, L2(o | u): the same recursion one level up              ← second-order pragmatic listener
p_attend = σ(b + w_pop·[u fits 1 object]                    ← capacity-limited attention
           − w_load·(objects − 2) − w_cand·(matches − 1))
P(o | u) = p_attend · L2(o | u)
         + (1 − p_attend) · capture(o)                      ← otherwise salience among the matching objects
if u fits all objects but one:
  with prob. p_out choose the one it does not fit           ← the lone odd-one-out
no informative word: P(o) = prior(o)
with prob. lapse choose uniformly
```

## Number game: is the target in the concept the examples come from?

`even_rule_digit_size_principle_counting`

```
sim(t) = share of examples ending in t's digit              ← graded last-digit similarity
       + s · (1 − pre·rule) · share of same parity          ← rule: the examples share a last digit
mutual = the same similarity among the examples
shortfall = softplus(mutual − sim(t))                       ← t judged against how alike the examples are
logit P(yes) = b + g·sim(t) − c·shortfall
  + w_f·[t a multiple of the examples' factor]              ← round factors (×5, ×10) weighted apart
  + w_m·(1 − pre_m·[any rule]) · exp(−gap / width)          ← decades outside the examples' range
                          any rule: a shared digit or factor ≥ 3
  + w_c·[t = k·x, x ≥ 3 an example] · exp(−cost·(k − 2))    ← counting up by multiples
  − w_s·(n − 2)·[t breaks the last-digit rule]              ← size principle
  − w_e·[all examples even, t odd]                          ← "even" is a privileged concept, "odd" is not
answer ~ Bernoulli(P(yes))
```